$x = [x] + \{x\}$ then ① $[x+y] \geq [x] + [y]$

integral fractional

② $[x+n] = [x] + n$

③ $[x/n] = [[x]/n]$

$x - 1 < [x] \leq x$

and $0 \leq \{x\} < 1$

$\because$ $x = \xi n + \cancel{r} + \cancel{\alpha}$

$[x] = \xi n + \cancel{r}$

- $[n/m]$ counts $1, \ldots, n$ divisible by m

- $\sum_{j=1}^{\infty} [n/p^j] =$ largest integer ℓ s.t. $p^\ell \mid n!$ $\leq \left[\frac{n}{p-1}\right]$

  $1 \ldots n$ divisible by p^j

  geometric series

- $\binom{m}{n} = \frac{m!}{n!\,(m-n)!}$ is integer $\because$ $\left[\frac{m}{p^j}\right] \geq \left[\frac{n}{p^j}\right] + \left[\frac{m-n}{p^j}\right]$

  numerator denominator

- $\frac{m!}{n_1! \cdots n_k!}$ is integer likewise

- multiplicative $f : \mathbb{N} \to \mathbb{R}$
  - $(m,n) = 1$ then $f(mn) = f(m) f(n)$
  - $f(1) = 1$ (otherwise vanish identically)
  - $n = p_1^{j_1} \cdots p_k^{j_k}$ then $f(n) = f(p_1^{j_1}) \cdots f(p_k^{j_k})$
  - $\sum_{d \mid n} f(d)$ is multiplicative $\because$ $\underbrace{\sum_{d \mid m} \sum_{d' \mid n}}_{(m,n)=1} \underbrace{f(d) f(d')}_{\text{multiplicative}}$

Euler's (totient) function $\phi(n)$

- (분모가 n인 기약분수) $\phi(n) = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right)$ "count $1, \ldots, n$ relatively prime to n"
- (분모가 n의 약수인 기약분수) $\sum_{d \mid n} \phi(d) = n$

$\frac{1}{6}$ $\frac{2}{6}$ $\frac{3}{6}$ $\frac{4}{6}$ $\frac{5}{6}$ $\frac{6}{6}$

$\phi(6)$ $\phi(3)$ $\phi(2)$ $\phi(3)$ $\phi(6)$ $\phi(1)$

$\phi(6) =$ (관계없이) $\mu(1) \cdot 6$

(2의 배수) $\mu(2) \cdot 3$

(3의 배수) $\mu(3) \cdot 2$

(6의 배수) $\mu(6) \cdot 1$ $= 2$

- let $n = p_1^{j_1} \cdots p_k^{j_k}$
- $\phi(n) = n \sum_{p \mid n} \left(1 - \frac{1}{p}\right)$

$$= n - \sum_{r} n/p_r + \sum_{r>s} n/(p_r p_s) - \sum_{r>s>t} n/(p_r p_s p_t)$$

$1 \ldots n$ $\quad$ $1 \ldots n$ divisible by p_r $\quad$ $1 \ldots n$ divisible by $p_r p_s$ $\quad$ $1 \ldots n$ divisible by $p_r p_s p_t$.

$$= \sum_{m=1}^{n} \underbrace{1 - \binom{l}{1} + \binom{l}{2} - \binom{l}{3} + \cdots}$$

$l(m)$: m을 나눌 수 있는 $p_1 \cdots p_k$

$$= \begin{cases} 1 & l = 0 \quad \text{i.e } m/n\text{은 기약분수} \\ 0 & \text{otherwise} \end{cases}$$

- ϕ is multiplicative also by properties of congruences
  - $\phi(p^j) = p^j - p^{j-1}$
  - $\phi(1) + \phi(p) + \cdots + \phi(p^j) = 1 + (p-1) + \cdots + (p^j - p^{j-1}) = p^j$

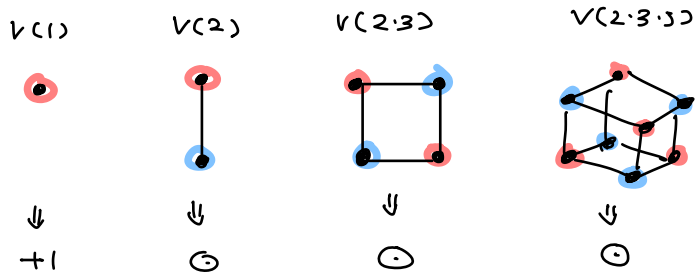
(Möbius Function) $\phi(n) = \sum_{d \mid n} \mu(d) \cdot (n/d)$ i.e 기약분수가 아닌 경 (n/d)개씩 조합으로 빼거나 더하기

Möbius Function $\mu(n)$

- $\mu(n)$ and $\nu(n)$ are multiplicative

- $\mu(n) = \begin{cases} 0 & \text{squared} \\ +1 & \text{even \# primes} \\ -1 & \text{odd \# primes} \end{cases}$

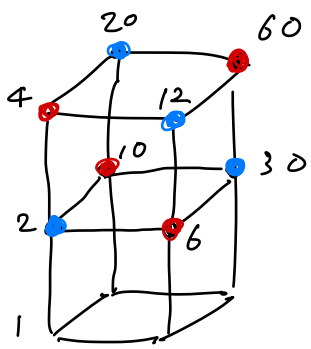
- $\nu(n) = \sum_{d|n} \mu(d) = \begin{cases} 1 & n=1 \\ 0 & n>1 \end{cases}$



Note. $\nu(p^j) = \underbrace{\mu(1)}_{+1} + \underbrace{\mu(p)}_{-1} + \underbrace{\mu(p^2) + \cdots}_{\text{square}}$

$= 0$

therefore $\nu(n) = 0 \quad (n>1)$



$$f(n) = g(n) - \sum_r g(n/p_r) + \sum_{r>s} g(n/p_r p_s) - \sum_{r>s>t} g(n/p_r p_s p_t)$$

$f(1)$	$f(2)$	$f(4)$
$f(2)$	$f(6)$	$f(12)$

Möbius Function은 수학과목을 위한 Ctrl-Z 스위치이다.

$$g(n) = \sum_{d|n} f(d) \iff f(n) = \sum_{d|n} \mu(d)\, g(n/d)$$

$$= \sum_{d'|n} \mu(n/d') \cdot g(d')$$

$\because \sum_{d|n} \mu(d) \cdot g(n/d)$

$= \sum_{d|n} \mu(d) \cdot \sum_{d'|n/d} f(d')$

$= \sum_{d'|n} f(d') \cdot \nu(n/d')$ $\quad n = d'$

$= f(n)$

$\because \sum_{d|n} f(d) = \sum_{d|n} f(n/d)$

$= \sum_{d|n} \sum_{d'|n/d} \mu(n/dd')\, g(d')$

$= \sum_{d'|n} g(d')\, \nu(n/d')$ $\quad n = d'$

$= g(n)$

$$g(x) = \sum_{n \le x} f(x/n)$$

then $f(x) = \sum_{n \le x} \mu(n) \cdot g(x/n)$

$\because \sum_{n \le x} \sum_{m \le x/n} \mu(n) \cdot f(x/mn)$

$= \sum_{l \le x} f(x/l) \cdot \nu(l) \quad (l = mn)$

5. $\tau(n)$, $\sigma(n)$

- number of divisors $\tau(n) = \prod_{p \mid n} (j+1)$ i.e $\sum_{d \mid n} 1$
- sum of divisors $\sigma(n) = \prod_{p \mid n} (1 + \cdots + p^j)$ i.e $\sum_{d \mid n} d$

- τ, σ multiplicative
  - $\tau(p^j) = j+1$
  - $\sigma(p^j) = 1 + p + \cdots + p^j$

- $\tau(n) < c n^{\delta}$ $(\delta > 0)$

$\because$ $f(n) = \tau(n)/n^{\delta}$ multiplicative.

$f(p^j) = \underbrace{(j+1)}_{\text{linear}} / \underbrace{p^{j\delta}}_{\text{exponential}} < 1$ except for $\underbrace{\text{finite } (j, p)\text{'s}}_{\text{select } c}$

- $\sigma(n) = \sum_{d \mid n} n/d \leq \sum_{d \leq n} n/d < n(1 + \log n)$

- this implies $\phi(n) > \frac{1}{4} n / \log n$ $(n > 1)$

$\because$
- $f(n) = \sigma(n)\phi(n)/n^2$ multiplicative
- $f(p^j) = \dfrac{p^{j+1} - 1}{p - 1} \, p^{j-1}(p-1) / p^{2j}$

  $= 1 - p^{-j-1} \geq 1 - 1/p^2$
- $\dfrac{\sigma(n)\phi(n)}{n^2} \geq \prod_{p \mid n} (1 - 1/p^2) \geq \prod_{m=2}^{\infty} (1 - 1/m^2) = 1/2$

6. Average Orders of τ, σ, ϕ

- τ : number of divisors

- $$\sum_{n \le x} \tau(\overset{\text{multiple}}{n}) = \sum_{d \le x} [x/\overset{\text{divisor}}{d}]$$

$$= x \log x + O(x) \qquad \because \sum_{d \le x} 1/d = \log x + O(1)$$

$$= x \log x + (2\gamma - 1)x \qquad (\text{Euler Constant})$$

$$+ O(\sqrt{x})$$

- average order : $\log n$ $\qquad \because \sum \log n \sim x \log x$.

- almost all numbers have $(\log n)^{\log 2}$ divisors

$\boxed{\text{proportion} \le x \text{ without property is } o(x)}$

i.e $\forall \varepsilon > 0$, almost all $n \in \mathbb{N}$

$$(\log n)^{-\varepsilon} < \tau(n)/(\log n)^{\log 2} < (\log n)^{\varepsilon}$$

- σ : sum of divisors

- $$\sum_{n \le x} \sigma(\overset{\text{multiple}}{n}) = \sum_{d \le x} \sum_{m \le x/d} m \qquad (dm = n)$$

$$= \sum_{d \le x} \frac{1}{2} [x/d]([x/d]+1)$$

$$= \sum_{d \le x} \frac{1}{2} (x/d)^2 + O(x/d)$$

$$= \frac{1}{12}\pi^2 x^2 + O(x \log x) \qquad \because \sum_{d \le x} 1/d^2 = \sum_{d=1}^{\infty} 1/d^2 + O(1/x)$$

- average order : $\frac{1}{6}\pi^2 n$ $\qquad \because \sum n \sim \frac{1}{2}x^2$

- ϕ : number of (p/q)'s in lowest terms

- $$\sum_{n\le x} \phi(n) = \sum_{n\le x}\sum_{d|n} \mu(d)\cdot(n/d)$$

$$= \sum_{d\le x} \mu(d) \sum_{m\le x/d} m$$

$$= \sum_{d\le x} \mu(d)\cdot\left(\tfrac{1}{2}(x/d)^2 + O(x/d)\right)$$

$$= (3/\pi^2)x^2 + O(x\log x) \qquad \because \sum_{d\le x}\mu(d)/d^2 = \sum_{d=1}^{\infty}\mu(d)/d^2 + O(1/x)$$

- average order : $6n/\pi^2$

- $Pr(p/q \text{ irreducible}) = 6/\pi^2$

$\approx 60\%$

$\because$
- $1 \le p \le q \le n$
- $n(n+1)/2$ pairs of p/q
- $\phi(1) + \cdots + \phi(n)$ pairs of p/q (lowest terms)

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7. Perfect Numbers

- n is perfect : $\sigma(n) = 2n$
  - $\sigma(6) = 1+2+3+6$ i.e $6 = 2^1 \cdot (2^2-1)$
  - $\sigma(28) = (1+2+4)(1+7) = 56$ i.e $28 = 2^2 \cdot (2^3-1)$

- we don't know if odd perfect number exists

- even number is perfect $\Longleftrightarrow$ $2^{p-1}(2^p-1)$ with p, 2^p-1 prime

$\ulcorner$ $(\Leftarrow)$
- let $n = 2^k \cdot m$ (n perfect ; m odd)
- then

$$\sigma(n) = 2n$$

$$(1+\cdots+2^k)\,\sigma(m) = 2 \cdot 2^k \cdot m$$

$$\sigma(m) = 2^{k+1} \cdot \underbrace{\frac{m}{2^{k+1}-1}}_{=:\,l}$$

- $m = (2^{k+1}-1)\cancel{l}$
- $\sigma(m) = (2^{k+1}\quad)\cancel{l} = (m+1)\cancel{l}$ then ① $l = 1$

  ② m is prime

$\Rightarrow$ $2^{k+1}-1$ is (Mersenne) prime

$\Rightarrow$ $(k+1)$ is prime

8. The Riemann zeta-function

- prime distribution

  related with zeta function $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s \quad (s = \sigma + it)$

Note. $\dfrac{1}{n^s} = \underbrace{\dfrac{1}{n^\sigma}}_{\text{크기}} \cdot \underbrace{\dfrac{1}{n^{it}}}_{\text{방향}}$ with $\left|\dfrac{1}{n^s}\right| = \dfrac{1}{n^\sigma}$

- $n^{it} = e^{it\log n} = \cos(t\log n) + i\sin(t\log n)$
- therefore $|n^{it}| = 1$

- $\sigma > 1$ then $\zeta(s)$ converges absolutely $\because \left|\dfrac{1}{n^s}\right| = \dfrac{1}{n^\sigma}$
- $\sigma > 1+\delta$ then $\zeta(s)$ converges uniformly $\because \dfrac{1}{n^\sigma} \le \dfrac{1}{n^{1+\delta}}$ (M-test)

- $\zeta(s)$ continued analytically throughout complex plane
- regular except for simple pole at $s=1$ with residue 1.

  i.e $\zeta(s) = \boxed{\dfrac{1}{s-1}} + c_0 + c_1(s-1) + c_2(s-1)^2 + \cdots$

- $\Xi(s) = \pi^{-\frac{1}{2}s}\,\Gamma(\tfrac{1}{2}s)\,\zeta(s)$ Note. $\Gamma(z) = \int_0^\infty t^{z-1}e^{-t}dt$
- $\Xi(s) = \Xi(1-s)$ $= (z-1)!$

---

claim $\zeta(s) = \dfrac{\pi^{s/2}\cdot\Xi(1-s)}{\Gamma(s/2)} = 0$ at $s = -2n$

$\because$ (해석적 연속) $\Gamma(z) = \Gamma(z+1)/z$

(무한대로 발산) $\Gamma(-n) = \dfrac{\Gamma(1)}{(-n)(-n+1)\cdots(-1)\,0}$

$$\sum_{n=1}^{\infty} 1/n^s = \prod_p (1-1/p^s)^{-1} \qquad (\sigma > 1)$$

zeta function Euler product

$\because$ • $\prod_{p \le N} (1-1/p^s)^{-1} = \prod_{p \le N} (1+p^{-s}+p^{-2s}+\cdots)$ (m : divisible only by primes $\le N$)

$= \sum_m m^{-s}$

• $\left| \sum_m m^{-s} - \sum_{n \le N} n^{-s} \right| \le \sum_{n > N} n^{-\sigma} \longrightarrow 0$ (as $N \to \infty$)

• $\sigma > 1$ then $\zeta(s)$ no zeros

• $\sigma < 0$ then $\zeta(s)$ only trivial zeros at $-2, -4, -6 \cdots$

• $0 \le \sigma \le 1$ then $\zeta(s)$ has

$\underbrace{\text{infinitely many}}_{\text{Hardy}}$ zeros $\underbrace{\text{(only?)}}_{\text{Riemann}}$ in $\sigma = \frac{1}{2}$

---

if Riemann Hypothesis true

① $\pi(x) = \int_2^x \frac{dt}{\log t} + O(\sqrt{x} \log x)$ and $p_{n+1} - p_n = O(p_n^{\frac{1}{2}+\varepsilon})$

Note. this is valid in weaker term

with narrow zero-free region left of line $\sigma = 1$

---

② $\sum_{n \le x} \mu(n) = O(x^{\frac{1}{2}+\varepsilon})$ because $1/\zeta(s) = \sum_{n=1}^{\infty} \mu(n)/n^s$

• $\sigma > 1$ then $\zeta(s) \cdot (1/\zeta(s)) = \sum_{n=1}^{\infty} \frac{1}{n^s} \nu(n)$

• vaild in $\sigma > \frac{1}{2}$ with Riemann Hypothesis

---

③ $\sigma > 2$ then $\zeta(s-1)/\zeta(s) = \sum_{n=1}^{\infty} \phi(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{n}{n^s}\right)\left(\sum_{d=1}^{\infty} \frac{\mu(d)}{d^s}\right)$

④ $\sigma > 1$ then $(\zeta(s))^2 = \sum_{n=1}^{\infty} \tau(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \cdot \left(\sum_{d=1}^{\infty} \frac{1}{d^s}\right)$

⑤ $\sigma > 2$ then $\zeta(s)\zeta(s-1) = \sum_{n=1}^{\infty} \sigma(n)/n^s$ $\because \left(\sum_{n=1}^{\infty} \frac{1}{n^s}\right) \cdot \left(\sum_{d=1}^{\infty} \frac{d}{d^s}\right)$